Crowded Video Individual Counting Informed by Social Grouping and Spatial-Temporal Displacement Priors

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***Abstract*—Video Individual Counting (VIC) is a recently introduced task aiming to estimate foot traffic from a video. It extends Video Crowd Counting (VCC) beyond the per-frame pedestrian count. In contrast to VCC that learns to count pedestrians across frames, VIC must identify co-existent pedestrians between frames, which turns out to be a correspondence problem. Existing VIC approaches, however, can underperform in congested scenes such as metro commuting. To address this, we build WuhanMetroCrowd, one of the first VIC datasets that characterize crowded, dynamic pedestrian flows. It features sparse-to-dense density levels, short-to-long video clips, slow-to-fast flow variations, front-to-back appearance changes, and light-to-heavy occlusions. To better adapt VIC approaches to crowds, we rethink the nature of VIC and recognize two informative priors: i) the social grouping prior that indicates pedestrians tend to gather in groups and ii) the spatial-temporal displacement prior that informs an individual cannot teleport physically. The former inspires us to relax the standard one-to-one (O2O) matching used by VIC to one-to-many (O2M) matching, implemented by an implicit context generator and an O2M matcher; the latter facilitates the design of a displacement prior injector, which strengthens not only O2M matching but also feature extraction and model training. These designs jointly form a novel and strong VIC baseline OMAN++. Extensive experiments show that OMAN++ not only outperforms state-of-the-art VIC baselines on the standard SenseCrowd, CroHD, and MovingDroneCrowd benchmarks, but also indicates a clear advantage in crowded scenes, with a 41.8% error reduction on our WuhanMetroCrowd dataset. Code, data, and pretrained models are available at https://github.com/tiny-smart/OMAN.**

***Index Terms*—Video individual counting, foot traffic, one-to-many matching, displacement priors.**

I. Introduction

MODERN urbanization and population growth have led to increasingly crowded public spaces, raising concerns about public security. Metro systems, as a representative, face significant challenges in managing congested foot traffic, especially during commuting hours [1]. Effective and efficient crowd analysis is therefore essential for optimizing traffic flows and enabling crowd management. While video surveillance and vision-based techniques such as Video Crowd Counting (VCC) [2]–[4] have been widely adopted to monitor crowds, VCC only addresses per-frame pedestrian counting, revealing deficiency in scenarios where the information of foot traffic within a period of time matters. To address this, cross-line crowd counting, a.k.a. Line of Interest (LOI) [5]–[7], and Multiple Object Tracking (MOT) [8]–[10] are introduced or repurposed to acquire such information. Yet, LOI cannot capture pedestrians of different walking directions, and the computational cost of MOT can increase quadratically with the number of pedestrians detected.

To better estimate foot traffic beyond LOI and MOT, a new task termed Video Individual Counting (VIC) is introduced recently [11]. As shown in Fig. 1(a), VIC aims to count

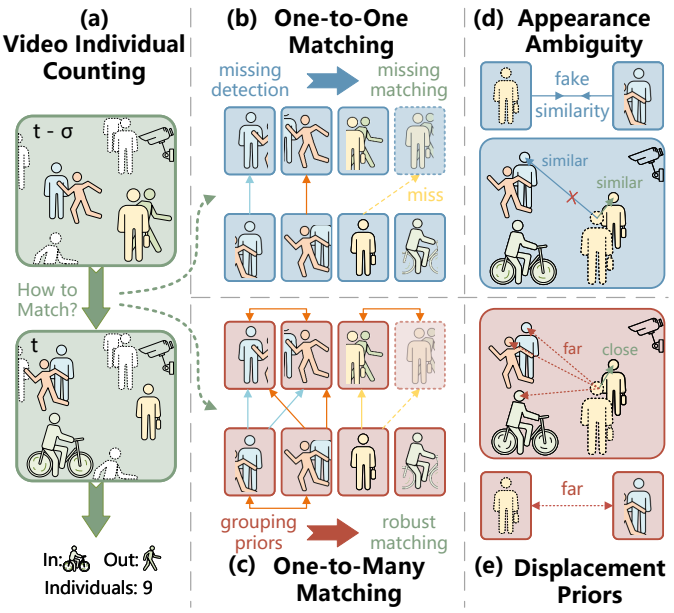


Fig. 1. **Video individual counting in crowded scenes**. (a) The key of VIC is to identify co-existent pedestrians between frames, but (b) existing methods typically apply an O2O matching strategy, suffering from missing detections due to occlusions (the yellow person at time t cannot match the undetected yellow person at time $t-\sigma$). (c) Inspired by the grouping prior of walking pedestrians, we relax the O2O matching to O2M matching that allows an individual to match a group to enhance matching robustness (the yellow person at time $t-\sigma$ is matched to the belonging group and is not counted as a new inflow). In addition, (d) matching informed by only appearance cues may not distinguish visually similar pedestrians, leading to wrong matches (the yellow dotted-line person at time $t-\sigma$ is falsely matched to the blue solid-line person at time t who is far away). In fact, (e) walking pedestrians obey spatial-temporal displacement prior that encourages spatially close matches (the yellow dotted-line person at time $t-\sigma$ tends to match who is nearby).

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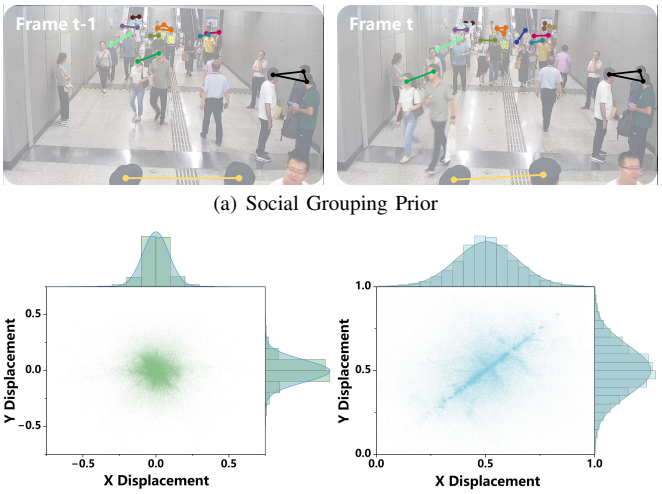
unique pedestrians across video frames to infer inflow/outflow numbers. While VIC still concerns counting, the essence of VIC turns out to be a correspondence problem, i.e., matching co-existent pedestrians between frames. Following this paradigm, a number of VIC approaches have been developed with increasingly improved performance [11]–[14]; however, we observe that they often underperform in crowded scenes. Such phenomena may stem from two facts: i) the deficiency of crowded VIC data and ii) the exact O2O matching conditioned on appearance only. Crowded scenes, which are common in VCC datasets, are less present in VIC datasets. A main reason seems that early VIC datasets are repurposed from MOT datasets [15], [16] such that the model does not see sufficient crowd patterns during training. In addition, crowded scenes typically characterize frequent occlusions and reduced appearance cues. The former can render wrong or missing pedestrian detections (Fig. 1(b)); the latter can raise great challenges for exact pedestrian matching, leading to wrong or missing matches (Fig. 1(d)).

To overcome the data dilemma, we introduce WuhanMetroCrowd, one of the first VIC datasets tailored to crowded scenes based on the Wuhan metro system. In particular, WuhanMetroCrowd hosts 80 surveillance videos from 15 metro stations collected during peak hours, holidays, and festivals, spanning from 2023 to 2025. As a result, it exhibits visual challenges including sparse-to-dense density levels, short-to-long video clips, slow-to-fast flow variations, front-to-back appearance changes, and light-to-heavy occlusions. Comprehensive manual annotations of $207,545$ pedestrians are also provided.

To alleviate the technical challenges in pedestrian matching, we rethink the nature of VIC and have made two important observations: i) per Fig. 2(a), pedestrians tend to gather by group, exhibiting a social grouping behavior; and ii) per Fig. 2(b), pedestrians probably move within certain ranges between consecutive frames. The former suggests a social grouping prior that encourages the use of *group context*; the latter implies a spatial-temporal displacement prior which reveals the fact that pedestrians cannot teleport physically and that pedestrian matching could be bounded locally.

In this work, we develop our approach informed by the two priors above. We first propose to relax the O2O matching into O2M matching to harness the group context. In contrast to O2O matching that penalizes all wrong matches (Fig. 1(b)), O2M matching can leverage shared appearance and context cues to enable individual-to-group assignments (Fig. 1(c)), which alleviates occlusions. Indeed the nature of the VIC problem tells us that *it is not necessary to match exactly the same pedestrian*. To exploit the displacement prior (Fig. 1(e)), we propose to incorporate it into the model to regularize both pedestrian matching and model training.

The O2M matching is implemented in a coarse-to-fine manner. In particular, an Implicit Context Generator (ICG) is introduced to infuse the group context into pedestrian descriptors. A One-to-Many Pairwise Matcher (OMPM) is then used to confirm the individual-to-group correspondence. This leads to a naive implementation termed OMAN. We then extend OMAN to OMAN++ by introducing a Displacement Prior Injector (DPI) to encapsulate the spatio-temporal displacement



(a) Social Grouping Prior

(b) Spatial-Temporal Displacement Prior

Fig. 2. **Two priors behind moving crowds**. (a) The social grouping prior indicating that pedestrians tend to walk in groups, with each social group connected with lines of different colors. (b) The spatial-temporal displacement prior revealing a limited range of pedestrian displacements (left) and their min-max normalized displacements (right) within three-second intervals, suggesting pedestrian motion is physically bounded.

prior via three synergistic modules: i) Displacement-Aware Self-Attention (DASA), which identifies geometrically plausible matches; ii) Displacement Modulator, which refines feature similarity; and iii) Displacement-informed Optimal Transport (D-OT) loss, which regularizes O2M matching. They work jointly to embed kinematic knowledge into the model.

Experiments on three public benchmarks, including SenseCrowd [15], CroHD [16] and MovingDroneCrowd [14], and our WuhanMetroCrowd dataset reveal that OMAN++ emerges to be a simple yet strong baseline for VIC. On the three public benchmarks, it outperforms state-of-the-art approaches without additional pretraining by $10.5\% \sim 34.7\%$ in MAE, $2.9\% \sim 20.9\%$ in MSE, and $5.5\% \sim 26.2\%$ in WRAE. Remarkably, OMAN++ reduces the MAE, MSE, and WRAE by 70.1%, 85.0% and 41.8%, respectively, on our WuhanMetroCrowd dataset, implying that our approach is dedicatedly optimized for crowded scenes. Ablation studies have also justified the contribution of each design. Aside from these results, we further study some practical issues such as the influence of the frame interval and the strategies used for incorporating the displacement prior. Our results suggest that *O2M matching is a more natural matching paradigm than O2O for VIC* and *the displacement prior is an instrumental cue for crowded VIC*.

Our contributions include the following.

- We suggest two informative VIC priors: the social grouping prior that describes grouped individuals and the spatial-temporal displacement prior that bounds inter-frame pedestrian motion.
- OMAN++: a simple and strong VIC baseline that implements O2M matching and motion-aware reasoning, which performs local individual-to-group matching by fusing appearance and motion cues, reducing the complexities of crowded VIC.
- WuhanMetroCrowd: A novel VIC benchmark that highlights crowded scenes, comprises 80 video sequences

with $11,925$ frames, and features long sequences, large flow and scale variations, and significant occlusions.

A preliminary conference version of this work appeared in [17]. We extend [17] by i) building a novel VIC benchmark WuhanMetroCrowd with significant data diversity and scene variations, which fills the data gap of crowded scenes and long sequences in VIC; ii) delving into the physical motion nature of VIC and identifying the spatial-temporal displacement prior; iii) introducing the cross-frame displacement-aware self-attention and a displacement-modulated O2M matcher; and iv) designing a displacement prior loss to enhance the learning of both appearance and motion representations. Besides limited experiments in [17], we now v) conduct comprehensive experiments on four different benchmarks with consistent and substantial improvements over the state of the art.

II. Related Work

To estimate foot traffic, early methods can be divided into two categories: *video crowd counting* and *multiple object tracking*; however, these methods struggle to provide precise inflow/outflow counts until a novel task called *video individual counting* is introduced.

A. Video Crowd Counting

Existing VCC methods can be divided into region-of-interest (ROI)-based VCC and line-of-interest (LOI)-based VCC. ROI-based methods aim to count per-frame pedestrians in videos. Early work [18], [19] treats crowd counting as a detection problem by converting point-level labels into pseudo bounding boxes for model training, where pedestrian counts are determined by the number of bounding boxes. Due to the limited performance of detection models in crowded scenes, some recent work [20], [21] introduces local constraints, but suffers from inaccurate pseudo bounding boxes. Thus, most this line of methods shift toward density map-based counting [22], [23]. Particularly, some work [2], [3] focuses on exploiting spatial-temporal information to improve VCC. Despite the success in accurate counting, density map-based approaches cannot localize crowds. To balance between counting and localization, localization-based methods [4], [24], [25] are proposed recently, aiming to directly predict the center points of heads. Generally, ROI-based counting only infers per-frame pedestrian counts rather than foot traffic.

LOI-based methods, a.k.a. cross-line crowd counting, count pedestrians who cross a pre-set line to estimate directed flux. Early work approaches the problem using different strategies: some work [26], [27] accumulates blobs or captures velocities by motion models before regressing the crossing-line counts, while others rely on the detect-and-track framework [28], [29]. Yet, these approaches either depend on motion segmentation or object detection, which struggles to tackle occlusions in crowds. To mitigate occlusions, temporal slices [5], [30] are introduced to directly regress counts on the line. However, the slices discard much spatial information such that these slice-based methods do not generalize to dense scenes as well. To count dense crowds, [6], [7] decouples the task into ROI counting and velocity estimation so that local counts can be accumulated around the LOI per velocity. However, since it is impractical to define LOI for each scenario in reality, LOI-based methods fail to count pedestrians of different directions.

In contrast to ROI-based methods that accumulate pedestrians in each frame and to LOI-based ones that only count those in certain directions, our methods directly estimate foot traffic across frames and over the entire image.

B. Multiple Object Tracking

Multiple Object Tracking (MOT), which simultaneously addresses localization, tracklet association, and re-identification (ReID), aims to track pedestrians with their identification (ID). Tracking by detection [8], [31]–[33] is the most widely adopted paradigm, which employs motion filters (e.g., Kalman filters) and IoU-based matching strategies to achieve detection and ReID; methods like HeadHunter-T [16] further promote such a paradigm to crowded scenes. These methods, however, suffer from limited detection performance in crowded scenes and from reliance on high-frame-rate tracklet associations. Another alternative is detection and tracking, where IDs are directly associated by similarity-aware regression or attention. JDE [34] first integrates object detection and appearance association into a unified model, enabling real-time MOT. FairMOT [35] and ChainedTracker [36] further generate detections and identity embedding jointly. Some other work [9], [37] directly operates on center points of heads to mitigate occlusions and scale variations. Recently, transformer-based trackers [10], [38]–[41] become dominant, implementing association by implicitly exploiting both spatial correspondence and temporal context.

Though these methods seem to function as a solution to VIC, the frequent ID switches—due to poor quality of features for detection and association—can cause errors on flux estimation. The high computational cost can also hinder practical applications, because trackers usually operate frame by frame at high rates. Instead, our approach excludes identity recognition and infers frames at an interval of $2 \sim 3$ seconds, avoiding potential ID mismatches and improving efficiency.

C. Video Individual Counting

In contrast to VCC and MOT, VIC estimates foot traffic in a distinct manner. DRNet [11], being the first work in this task, formulates VIC as a problem of pedestrian matching and predicts both pedestrian inflows and outflows. While it outperforms MOT approaches, it still follows the problem setup of MOT and uses the identity labels of individuals. With a deeper insight that VIC is about matching rather than tracking, CGNet [12] replaces identity annotations with categorical ones and proposes a group-level matching loss for supervision. To execute matching, both DRNet [11] and CGNet [12] adopt the idea of explicit O2O matching such as Hungarian matching [42] and optimal transport [43] to identify co-existent pedestrians between adjacent frames. In this way, empirical thresholds are often required to execute bipartite matching, leading to parameter sensitivity. Alternatively, FMDC [44] and PDTR [13] exploit regression to match the same pedestrian between frames, but they still rely on



Fig. 3. **Overview of WuhanMetroCrowd dataset**. This map shows the geographical distribution of data collection, and annotation number of each station.

O2O correspondence to associate pedestrians. More recently, SDNet [14] integrates matching and counting using cross-frame cross-attention to search for homologous appearance, and VIC-SSL [45] exploits additional self-supervised pre-training to boost VIC representations based on crowd flows. Notably, these regression-based methods rely on Gaussian-like density maps, rendering difficulty in establishing point-to-point correspondences between pedestrians.

Instead of executing exact O2O matching, we propose to use O2M matching to facilitate the use of social grouping nature and contextual cues between pedestrians, improving robustness under occlusions and crowded scenes. To alleviate the reliance on varied appearance, we further exploit motion cues like displacement to inform our model with spatial-temporal context.

III. WuhanMetroCrowd Dataset

Existing VIC datasets primarily focus on short video clips with sparse or moderate crowd densities, where they fail to reflect the real-world challenges within long intervals and in crowded scenes, overlooking critical aspects such as accumulated errors, extreme density variations, severe occlusions, and large perspective changes. To fill this gap, we introduce a novel, large-scale dataset, termed WuhanMetroCrowd, tailored to dense VIC from a surveillance view, featuring diverse crowd dynamics and extended video clips. Particularly, WuhanMetroCrowd highlights high-density scenarios, including densely packed crowds, persistent occlusions, and abrupt scale shifts, providing a thorough benchmark for evaluating VIC models in real-world scenarios. To our knowledge, WuhanMetroCrowd is the first surveillance-perspective, crowded, and VIC-orientated dataset focusing on metro transport. Details and properties of the dataset are illustrated in what follows.

A. Data Collection, Annotation, and Partition

a) Data Collection: We collect 80 surveillance videos from 15 metro stations from the Wuhan metro network, as shown in Fig. 3. These stations include transfer stations, train stations, commercial districts, and other busy traffic hubs. $11,925$ frames were extracted with two-second sampling intervals from the 80 videos. The longest video sequence is of 864 frames. Video data were collected from 2023 to 2025, with a particular focus on densely crowded scenes including peak hours, holidays, and festivals. The video resolutions of our dataset are of 720×576, 1280×720, and 1920×1080. Fig. 4 exhibits 7 representative scenarios encompassed in our dataset: platform, transfer, fare gate, escalator, security, exit/entrance, and lobby, covering both sparse and dense scenes.

b) Data Annotations: We allocate 10 well-trained annotators and 2 checkers to conduct data annotation. The annotation process is fully manual without automatic detection or pre-labeling, and X-AnyLabeling[1] is used to annotate each pedestrian head and its inflow and outflow label. As shown in Fig. 4, by comparing three adjacent frames when labeling the intermediate (current) frame, $207,545$ pedestrians were annotated with four types of labels: i) pedestrians that co-exist in the current and next frame are labeled as *pedestrian*; ii) pedestrians that disappear in the next frame are labeled as *outflow*; iii) pedestrians that do not appear in the previous frame are labeled as *inflow*; iv) pedestrians that are absent in both the previous and the next frames—i.e., indicating simultaneous inflow and outflow events—are labeled as *both*, which typically occurs when a pedestrian (usually walking fast or at the border of the field of view), swiftly enters and exits the frame between two consecutive frames within the 2-second sampling interval. Additionally, we also provide

[1]https://github.com/CVHub520/X-AnyLabeling

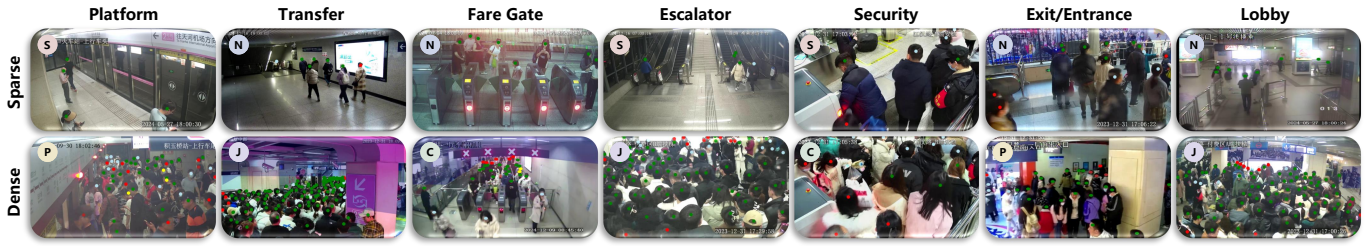
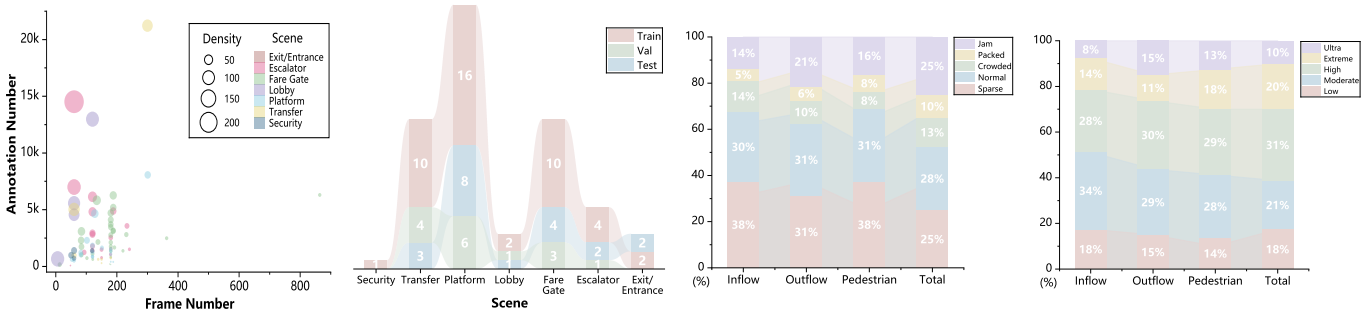


Fig. 4. **Annotated samples from WuhanMetroCrowd dataset**. Scenes are shown on the top, and some relatively sparse scenes are shown in the first row while dense scenes in the second row. Green, red, light blue, and yellow points respectively represent the class 'pedestrian', 'inflow', 'outflow', and 'both', and transparent dark blue polygon regions represent masks. 'S', 'N', 'C', 'P', and 'J' on the left-top of each picture, separately represents 5 different density levels, 'Sparse', 'Normal', 'Crowded', 'Packed', and 'Jam'.

TABLE I

COMPARISON WITH OTHER DATASETS FOR VIC. *#Frm./Seq.* REPRESENTS FRAMES PER VIDEO SEQUENCE WITH σ-FRAME INTERVALS, AND *#Relative Flow Variation per Clip* REPRESENTS THE AVERAGE OF THE FLOW VARIATION VALUES OF PEOPLE WITH DIFFERENT LABELS ACROSS ALL VIDEO CLIPS. '-' IS DUE TO UNRELEASED DATA.

Dataset	Task	Frames	Annotations	σ	Label	#Frm./Seq.	View	#Relative Flow Variation on Average			
								Inflow	Outflow	Pedestrian	Total
SenseCrowd [15]	MOT	62,938	2,344,276	15 (3s)	ID	6.62	surveillance	0.31	0.13	0.28	0.25
CroHD [16]	MOT	11,464	2,275,286	75 (3s)	ID	16.98	drone	0.03	0.03	0.46	0.46
UAVVIC [12]	VIC	5,396	398,158	1 (3s)	In/Out	24.42	drone	-	-	-	-
MovingDroneCrowd [14]	VIC	4,940	325,542	4 (-)	ID	13.88	drone	0.31	0.31	0.56	0.64
WuhanMetroCrowd (Ours)	VIC	11,925	207,545	1 (2s)	In/Out	149.06	surveillance	0.77	0.86	1.85	2.36



(a) Statistics w.r.t. Sequences (b) Dataset Partition (c) Density Variation (d) Flow Variation

Fig. 5. **Density and variation distribution of WuhanMetroCrowd dataset**. (a) depicts the statistical overview of our datasets. Each point denotes a video clip, X-axis represents frame number it contains, Y-axis counts its total annotation number, its scale represents the average density across the video, and the color denotes its belonging scene. (b) shows the dataset partition w.r.t scenes. (c) split class 'inflow', 'outflow', 'pedestrian', and total counts into five density levels. (d) similarly split them into five variation levels, illustrating that a single scene of WuhanMetroCrowd dataset may cover both static and dynamic situation.

masks for severely occluded and excessively blurred regions where identities of pedestrians are completely indiscernible. Specifically, mask regions are manually drawn as polygons; both ground-truth and predicted points falling within a mask region are excluded from evaluation. The density distribution of WuhanMetroCrowd dataset is shown in Fig. 5(a), which indicates the relation among annotation number, frame number, crowd density, and scenarios.

c) Data Partition: To ensure data diversity and mitigate potential biases, we split the dataset (Fig. 5(b)) into 45, 15 and 20 video sequences separately for training (60%), validation (20%), and testing (20%) with the following criteria in mind: i) different subsets are disjoint, covering different video sequences to avoid exactly the same scene between subsets; ii) each subset covers different scene types and video length; and iii) each subset covers sufficiently large density variations and flow variations. Note that some scene types (e.g., Security) have relatively few video sequences in total; it is therefore impractical to assign every scene type to every subset while maintaining the 60/20/20 split. For such rare scenes, we prioritise their inclusion in the training set to ensure the sufficient data diversity of training, accepting that some scene types may be absent from validation or test subsets.

B. Benchmark Properties and Statistics

a) Extended Long Video Clips: While VIC targets cumulative pedestrian inflows within a period of time, the inflow errors also accumulate with time. However, existing datasets, particularly those repurposed from MOT benchmarks [15], [16], mostly host short video clips of tens of seconds (Table I). Such temporal truncation creates two limitations: i) accumulated errors are small and fail to reflect the long-

term counting capability of the model, and ii) models trained on few-frame sequences can have biased representations and may perform poorly in scenarios requiring sustained temporal reasoning. To overcome these limitations, most videos in our WuhanMetroCrowd dataset are clipped into $60 \sim 240$ frames (Fig. 5(a)), i.e., $2 \sim 8$ minutes.

b) Significant Density and Flow Variations: We characterize density and flow variations with five levels, respectively. Fig. 5(c) shows five density levels of the three annotated categories and total counts. Considering the scales of pedestrians due to camera angle and imaging height, for 'inflow' and 'outflow', we split the average density (average counts across frames) per sequence into $\texttt{sparse} \in (0, 1]$, $\texttt{normal} \in (1, 2]$, $\texttt{crowded} \in (2, 3]$, $\texttt{packed} \in (3, 4]$, and $\texttt{jam} \in (4, +\infty]$ with an interval of 1, and set the interval to 5 for 'pedestrian' and 10 for total counts. Fig. 4 also provides some cases of these 5 different density levels. Notably, the *densest* sequence on average contains 21 inflows, 12 outflows, 230 pedestrians, and 242 total counts. Considering that some relatively static frames may decrease the average density, we depict the flow variation range of 3 classes and total counts in each clip to capture the temporal flow variation in each scenario, as shown in Fig. 5(d). The interval is 4 for 'inflow' and 'outflow', 10 for 'pedestrian', and 12 for total counts. It can be observed that our dataset provides exhaustive coverage of flow variations, ranging from sparse to extremely crowded scenarios. Particularly, the *maximum* flow variation range is 28 for 'inflow', 28 for 'outflow', 134 for 'pedestrian', and 144 for total counts. The relative flow variation on average (relative to total pedestrian counts) in Table I also highlights the dynamic nature of our dataset, which demonstrates that both sparse and dense crowds can be temporally included in one single clip of our dataset.

c) Severe Occlusions and Scale Variations: Since VIC still falls in the generic domain of crowd counting, VIC approaches should be robust to occlusions and scale variations commonly present in crowd counting. However, existing VIC data [15], [16] are mostly captured from nearly bird-eye views; the overlap between individuals is minimal and target scales remain consistent, making even the most crowded scenes insufficiently challenging for VIC. Considering the characteristics of surveillance cameras, which typically feature lower heights and smaller viewing angles compared to drone perspectives, occlusion and scale variations are particularly more prevalent in our dataset than in others. Fig. 4 provides some annotated examples of dense scenarios, where people are severely occluded and their scales vary.

d) Potential Use by Other Fields: Beyond VIC, WuhanMetroCrowd is also suited for i) video crowd counting, given its dense, long-sequence annotations; ii) crowd density estimation and anomaly detection, for its diversity of density levels and scene types; and iii) potential pedestrian flow analysis for urban transport planning under real-world metro context.

IV. Proposed Method

The social grouping prior forms robust appearance, hedging against occlusion; the spatial-temporal displacement prior encodes the physical constraint, suppressing geometrically implausible matches. The two priors are conceptually composed in a Bayesian fashion. Let $\mathcal{M}_{ij} \in \{0, 1\}$ indicate whether pedestrian j at time t matches pedestrian i at time $t-1$, $\boldsymbol{f}_i, \boldsymbol{f}_j$ denote their appearance features, and $\boldsymbol{\Delta}_{ij}$ their spatial displacement. For compactness, we write $M_{ij} \equiv [\mathcal{M}_{ij} = 1]$ to denote the event of a positive match. Under a conditional independence assumption on appearance and displacement given the match state, the posterior factorizes as

$$p(M_{ij} \mid \boldsymbol{f}_i, \boldsymbol{f}_j, \boldsymbol{\Delta}_{ij}) \propto p(\boldsymbol{f}_i, \boldsymbol{f}_j \mid M_{ij}) \cdot p(\boldsymbol{\Delta}_{ij} \mid M_{ij}), \quad (1)$$

where appearance features and spatial displacement contribute as independent likelihood terms, and are instantiated by the subsequent modules.

A. Problem Formulation

Given a video sequence $I = \{I_0, I_1, ..., I_T\}$ with $T+1$ frames, where the t-th frame I_t contains N_t pedestrians, the video is sampled with an interval of σ frames for efficiency. By coupling adjacent frames as pairs $\{I_{(k-1)\sigma}, I_{k\sigma}\}_{k=1}^{T/\sigma}$, in the k-th frame pair, *pedestrians* who do not exist in $I_{(k-1)\sigma}$, but show up in $I_{k\sigma}$ for the first time, are denoted as *inflows*, with the inflow number $N_{k\sigma}^{\texttt{in}}$. In contrast, *pedestrians* visible in $I_{(k-1)\sigma}$ but disappeared in $I_{k\sigma}$ are regarded as *outflows*, with the outflow number $N_{(k-1)\sigma}^{\texttt{out}}$. The goal of VIC is to count unique pedestrians from a video sequence, which can be formulated by

$$N_{\texttt{total}} = N_0 + \sum_{k=1}^{\texttt{T}/\sigma} N_{k\sigma}^{\texttt{in}}, \quad (2)$$

where $N_{\texttt{total}}$ is the total number of unique pedestrians, and N_0 is the first-frame pedestrian count.

For ease of exposition, we use t and $t-1$ to indicate the timestamp with an interval of σ in what follows.

B. Overview

The technical pipeline of OMAN++ is shown in Fig. 6. Following CGNet [12], given pedestrian locations inferred by a locator, we first crop pedestrian patches of size $h \times w$ from video frames and compute cross-frame location displacements between pedestrians. The pedestrian features are then extracted with position embedding, followed by an Implicit Context Generator (ICG) used to implicitly provide the group context to inform potential pedestrian matches via multi-head decoupled self-attention. Specifically, OMAN (our preliminary conference baseline [17]) uses standard Multi-Head Self-Attention (MHSA) blocks, while OMAN++ replaces these with Displacement-Aware Self-Attention (DASA) blocks as part of the Displacement Prior Injector described in Section IV-E. A One-to-Many Pairwise Matcher (OMPM) is used to identify individual-to-group correspondences. To exploit the displacement prior, a Displacement Prior Injector (DPI) can work through a Displacement-Aware Self-Attention (DASA) block (in place of the multi-head self-attention block), a displacement modulator, and an improved Optimal Transport (OT) loss. In what follows, we first define the problem setting and then explain the technical details.

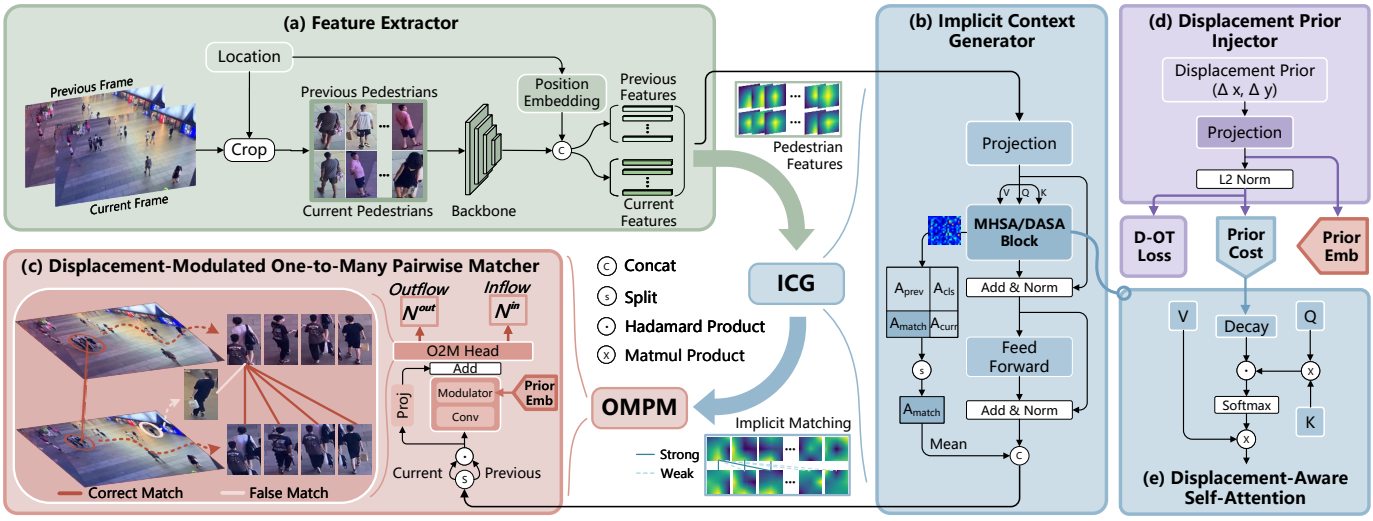


Fig. 6. **Technical pipeline of OMAN++.** (a) Individuals are first located and cropped with pre-provided location, followed by position-embedded feature extraction. We then apply (b) Implicit Context Generator (ICG) to execute implicit context enhancement, and resort to (c) One-to-Many Pairwise Matching (OMPM) to implement O2M assignment via a matching head, generating individual-to-group matches and therefore the inflow and outflow numbers. To capture motion patterns of pedestrians, we further devise (d) Displacement Prior Injector (DPI) to inject displacement prior into each module through (e) Displacement-Aware Self-Attention (DASA), a displacement modulator, and a displacement-informed OT (D-OT) loss, keeping final inflow and outflow predictions under the physical constraint.

C. Implicit Context Generator

Inspired by a recent observation [46] that self-attention can be decoupled to capture the implicit token relations for similarity matching, we introduce an ICG module to generate the group context and build the pedestrian relations between frames to inform potential pedestrian matches.

Formally, let $m = N_{t-1}$ and $n = N_t$ denote the pedestrian count in the $(t-1)$-th and the t-th frames, respectively. As shown in Fig. 6(b), the encoded pedestrian descriptors are first reshaped and projected into a d-dimensional feature vector, denoted by $\boldsymbol{F}_t \in \mathbb{R}^{n\times d}$. By concatenating $\boldsymbol{F}_{t-1}$ and $\boldsymbol{F}_t$, we arrive at $\boldsymbol{F}_{t-1,t} \in \mathbb{R}^{(n+m)\times d}$, which is then fed to the standard multi-head self-attention blocks. After self-attention, an attention map $\boldsymbol{A} \in \mathbb{R}^{(n+m)\times(n+m)}$, which indicates the affinities between tokens, can be extracted and split into four sub-attention maps [46], namely $\boldsymbol{A}_{\texttt{prev}} \in \mathbb{R}^{m\times m}$, $\boldsymbol{A}_{\texttt{cls}} \in \mathbb{R}^{m\times n}$, $\boldsymbol{A}_{\texttt{match}} \in \mathbb{R}^{n\times m}$, and $\boldsymbol{A}_{\texttt{curr}} \in \mathbb{R}^{n\times n}$, taking the form

$$\boldsymbol{A} = \begin{bmatrix} \boldsymbol{A}_{\texttt{prev}} & \boldsymbol{A}_{\texttt{cls}} \\ \boldsymbol{A}_{\texttt{match}} & \boldsymbol{A}_{\texttt{curr}} \end{bmatrix}_{(n+m)\times(n+m)}, \tag{3}$$

where $\boldsymbol{A}_{\texttt{prev}}$ and $\boldsymbol{A}_{\texttt{curr}}$ indicate the pedestrian similarity of the previous frame ($\boldsymbol{F}_{t-1}$) and of the current frame ($\boldsymbol{F}_t$), respectively. $\boldsymbol{A}_{\texttt{match}}$ captures the attention from current-frame pedestrians to previous-frame pedestrians (candidate matches), while $\boldsymbol{A}_{\texttt{cls}}$ does from a reverse direction. Both sub-maps are column-averaged to produce scalar context scores per pedestrian, and their concatenation provides richer bidirectional group context. To characterize the group context, we average $\boldsymbol{A}_{\texttt{match}}$ and $A_{\texttt{cls}}$ by column to derive $\bar{\boldsymbol{A}}_{\texttt{match}} \in \mathbb{R}^n$ and $\bar{\boldsymbol{A}}_{\texttt{cls}} \in \mathbb{R}^m$. By concatenating both $\bar{\boldsymbol{A}}_{\texttt{match}}$ and $\bar{\boldsymbol{A}}_{\texttt{cls}}$ with the output of self-attention, we acquire the d'-dimensional context-informed representation $\boldsymbol{F}'_{t-1,t} \in \mathbb{R}^{(n+m)\times d'}$ indicating the degree of cross-frame pedestrian connections, that is,

$$\boldsymbol{F}'_{t-1,t} = \texttt{g}\left(\texttt{attn}(\boldsymbol{F}_{t-1,t}) \oplus (\bar{\boldsymbol{A}}_{\texttt{match}} \oplus \bar{\boldsymbol{A}}_{\texttt{cls}})\right), \tag{4}$$

where $\oplus$ is the concatenation operator, $\texttt{attn}(\cdot)$ is the multi-head self-attention block, and $\texttt{g}(\cdot)$ is a feed-forward layer with residual connections.

D. One-to-Many Pairwise Matcher

Here we introduce the OMPM implementing the O2M assignment. By assigning an individual to a group, one can harness the social grouping prior of pedestrians, especially in crowded scenes. This benefit is also mentioned in some related tasks such as ReID [47] and MOT [10].

We formulate the O2M assignment as a classification problem. Concretely, the context-enhanced cross-frame pedestrian representation $\boldsymbol{F}'_{t-1,t}$ is first split into single-frame representations $\boldsymbol{F}'_{t-1} = \{\boldsymbol{f}^1_{t-1}, \boldsymbol{f}^2_{t-1}, \ldots, \boldsymbol{f}^m_{t-1}\} \in \mathbb{R}^{m\times d'}$ and $\boldsymbol{F}'_t = \{\boldsymbol{f}^1_t, \boldsymbol{f}^2_t, \ldots, \boldsymbol{f}^n_t\} \in \mathbb{R}^{n\times d'}$, where $\boldsymbol{f}^i_{t-1} \in \mathbb{R}^{d'}$ corresponds to the i-th pedestrian in the $(t-1)$-th frame, $i = 1, ..., m$, and $\boldsymbol{f}^j_t \in \mathbb{R}^{d'}$ the j-th pedestrian in the t-th frame, $j = 1, ..., n$. Then the similarity vector $\boldsymbol{s}^{ij}_{t-1,t} \in \mathbb{R}^{d'}$ is computed as $\boldsymbol{s}^{ij}_{t-1,t} = \boldsymbol{f}^i_{t-1} \odot \boldsymbol{f}^j_t$, where $\odot$ is the Hadamard product. Finally, $\boldsymbol{s}^{ij}_{t-1,t}$ is fed into an MLP to generate the probability of a candidate match $p^{ij}_{t-1,t}$ using a $\texttt{sigmoid}$ function by

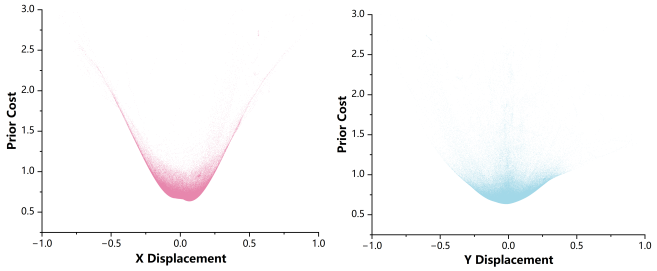
$$p^{ij}_{t-1,t} = \texttt{sigmoid}\left(\texttt{MLP}(\boldsymbol{s}^{ij}_{t-1,t})\right). \tag{5}$$

In this way, the inflow count $N^{\texttt{in}}_t$ of the t-th frame and the outflow count $N^{\texttt{out}}_{t-1}$ of the $(t-1)$-th frame can be inferred by

$$N^{\texttt{in}}_t = N_t - \sum_{j=1}^{n} \left\lfloor \max_{i=1}^{m} p^{ij}_{t-1,t} \right\rceil, \tag{6}$$

$$N^{\texttt{out}}_{t-1} = N_{t-1} - \sum_{i=1}^{m} \left\lfloor \max_{j=1}^{n} p^{ij}_{t-1,t} \right\rceil, \tag{7}$$

where $\lfloor\cdot\rceil$ rounds to the nearest integer.



(a) Prior Cost w.r.t X Displacement (b) Prior Cost w.r.t Y Displacement

Fig. 7. **Relationship between the prior cost and the displacement.** Each point denotes a matching pair. The pedestrian displacements are separately shown along the (a) x-axis and (b) y-axis. The prior cost is generally positively correlated to the absolute value of displacement in both x and y direction, while the trends differ that the prior cost increase faster when pedestrians in frames move horizontally than move vertically.

During inference, a pedestrian in the t-th frame is allowed to match a different one in the $(t-1)$-th frame, as long as they are treated as a group. Conceptually, the differences between the constraints for O2O and O2M matching are

$$\begin{aligned} &\texttt{O2O}: \quad \boldsymbol{M}\mathbb{1}_m \leq 1, \quad \boldsymbol{M}^T\mathbb{1}_n \leq 1 \\ &\texttt{O2M}: \quad \boldsymbol{M}\mathbb{1}_m \leq \eta \end{aligned}, \tag{8}$$

where $\boldsymbol{M} \in \mathbb{R}^{n\times m}$ is the matching matrix, $\mathbb{1}_n$ and $\mathbb{1}_m$ are n-dimensional and m-dimensional column vectors of ones, and η is the maximum size of a group.

E. Displacement Prior Injector

In the vanilla design of ICG and OMPM, only appearance features are considered to implement O2M matching. However, appearance cues can degrade in crowded scenes. Here we propose to incorporate the displacement prior into the pipeline to facilitate robust matching.

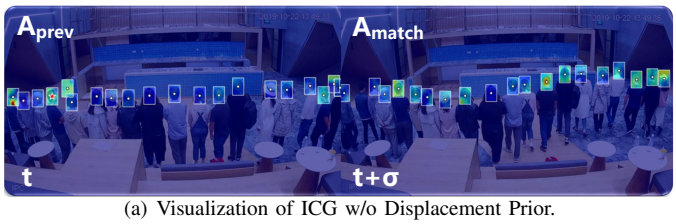
To exploit the prior, as shown in Fig. 6(d), the DPI is introduced to guide O2M matching. Specifically, let pedestrian locations in the $(t-1)$-th and the t-th frame be denoted by two point sets $\boldsymbol{X}_{t-1} \in \mathbb{R}^{m\times 2}$ and $\boldsymbol{X}_t \in \mathbb{R}^{n\times 2}$, respectively. We concatenate them as $\boldsymbol{X}_{t-1,t} = [\boldsymbol{X}_{t-1}; \boldsymbol{X}_t] \in \mathbb{R}^{(n+m)\times 2}$ and compute the pairwise displacement $\mathcal{D} \in \mathbb{R}^{(n+m)\times(n+m)\times 2}$ by

$$\mathcal{D}^{ij} = \boldsymbol{X}^j_{t-1,t} - \boldsymbol{X}^i_{t-1,t}, \ i,j = 1,...,n+m\,. \tag{9}$$

A nonlinear projection $\phi(\cdot)$ generates the displacement prior embedding $\mathcal{P} \in \mathbb{R}^{(n+m)\times(n+m)\times\frac{d}{hw}}$, with $\mathcal{P} = \phi(\mathcal{D})$.

In what follows, we show through three technical improvements on how to use the prior embedding $\mathcal{P}$ to improve representation, matching, and learning, respectively.

a) Displacement-Aware Self-Attention: In Section IV-C, we simply leverage a standard MHSA block. Inspired by [48], here we introduce a novel displacement-aware self-attention (DASA) block to incorporate motion cues. In the ICG module, the pedestrian representation $\boldsymbol{F}_{t-1,t}$ is normalized and projected to the query $\boldsymbol{Q} \in \mathbb{R}^{(n+m)\times d}$, key $\boldsymbol{K} \in \mathbb{R}^{(n+m)\times d}$ and value $\boldsymbol{V} \in \mathbb{R}^{(n+m)\times d}$, and are then fed into DASA in Fig. 6(e). The prior embedding $\mathcal{P}$ is then used to compute



(a) Visualization of ICG w/o Displacement Prior.



(b) Visualization of ICG w/ Displacement Prior.

Fig. 8. **Visualization of displacement guided attention.** (a) shows the power of vanilla self-attention, and (b) depicts how we apply displacement prior to guide self-attention, where white boxes are crop patches with attention map inside. The pedestrian with red star in t-th frame serves as a candidate to search in the $(t+\sigma)$-th frame.

the prior cost $\boldsymbol{\gamma} \in \mathbb{R}^{(n+m)\times(n+m)}$, modulating the query-key product via the Hadamard product, which takes the form

$$\begin{aligned} &\boldsymbol{Q} = \boldsymbol{W}_Q\boldsymbol{F}_{t-1,t}\,, \quad \boldsymbol{K} = \boldsymbol{W}_K\boldsymbol{F}_{t-1,t}\,, \\ &\boldsymbol{V} = \boldsymbol{W}_V\boldsymbol{F}_{t-1,t}\,, \quad \boldsymbol{\gamma} = \|\mathcal{P}\|_2\,, \\ &\texttt{DASA}(\boldsymbol{F}_{t-1,t}) = \left(\texttt{Softmax}(\frac{\boldsymbol{Q}\cdot\boldsymbol{K}^\top}{\sqrt{d}}) \odot \theta^{\boldsymbol{\gamma}}\right)\boldsymbol{V}\,, \end{aligned} \tag{10}$$

where $\boldsymbol{W}_Q$, $\boldsymbol{W}_K$ and $\boldsymbol{W}_V$ are learnable parameters, $\|\cdot\|_2$ is the ℓ_2 norm across all channels, $\texttt{Softmax}(\cdot)$ is the softmax function, and θ is a learnable parameter. To confirm the difference between the prior cost and the displacement itself in both directions and amplitudes, we explore their relationship in Fig. 7, showing two clear but different quadratic dependences of displacements along the two axes.

We also compare the attention map between MHSA and DASA in Fig. 8 to highlight the guidance of the displacement prior. The ICG module fetches $\boldsymbol{F}'_{t-1,t}$ and provides implicit context for OMPM. With DASA, Eq. (4) amounts to

$$\boldsymbol{F}'_{t-1,t} = \texttt{g}\left(\texttt{DASA}(\boldsymbol{F}_{t-1,t}) \oplus (\bar{\boldsymbol{A}}_{\texttt{match}} \oplus \bar{\boldsymbol{A}}_{\texttt{cls}})\right)\,, \tag{11}$$

where $\texttt{DASA}(\cdot)$ is our proposed DASA block, and $\texttt{g}(\cdot)$ is a feed-forward layer with residual connections.

b) Displacement-Modulated Matcher: For dense crowds, pedestrian motion is usually suppressed. Thus, the background and group context should be emphasized for matching stability, and vice versa for sparse crowds. Consequently, we introduce a coarse-to-fine modulator in Fig. 6(c) to adaptively weight the contributions of appearance features and motion-based geometric cues, superseding the standard MLP in Section IV-D.

Given the similarity representation $\boldsymbol{s}^{ij}_{t-1,t}$, a linear projection $\delta(\cdot)$ and layer normalization $\texttt{LN}(\cdot)$ are first applied to form a representation $\boldsymbol{z}_{\texttt{coarse}} \in \mathbb{R}^{\frac{d}{hw}}$ specific to coarse modulation, which captures global similarity but lacks fine-grained spatial details with explicit structural modeling, defined by

$$\boldsymbol{z}_{\texttt{coarse}} = \texttt{LN}\left(\delta(\boldsymbol{s}^{ij}_{t-1,t})\right)\,. \tag{12}$$

In addition, we reshape $\boldsymbol{s}^{ij}_{t-1,t}$ to a 3D tensor $\mathcal{S}^{ij}_{t-1,t} \in \mathbb{R}^{h\times w\times \frac{d'}{hw}}$ to restore the local spatial context. To incorporate the displacement prior, for a cross-frame pedestrian pair (i, j), we extract the corresponding slice $\mathcal{P}^{ij} \in \mathbb{R}^{\frac{d}{hw}}$ from the cross-frame block of $\mathcal{P}$, broadcast it to $\mathcal{P}' \in \mathbb{R}^{h\times w\times \frac{d}{hw}}$ by repeating along the spatial dimensions, use $\mathcal{P}'$ to modulate the $(\frac{d}{hw})$-channel convolutional response of $\mathcal{S}^{ij}_{t-1,t}$, and generate another representation $\boldsymbol{z}_{\texttt{fine}} \in \mathbb{R}^{h\times w\times \frac{d}{hw}}$ w.r.t. fine modulation, allowing the similarity map to adapt its response conditioned on the displacement. This procedure can be formulated by

$$\boldsymbol{z}_{\texttt{fine}} = \alpha(\mathcal{P}') \odot \texttt{ReLU}\left(\texttt{Conv}(\mathcal{S}^{ij}_{t-1,t})\right) + \beta(\mathcal{P}')\,, \tag{13}$$

where $\texttt{Conv}(\cdot)$ denotes a convolution layer, $\alpha(\cdot)$ and $\beta(\cdot)$ are linear layers, and $\odot$ is the Hadamard product, $\texttt{ReLU}$ is the rectified linear unit activation function. Then we obtain the fused representation $\boldsymbol{z}_{\texttt{fused}}$ by

$$\boldsymbol{z}_{\texttt{fused}} = \boldsymbol{z}_{\texttt{coarse}} + \texttt{GAP}(\boldsymbol{z}_{\texttt{fine}})\,, \tag{14}$$

where $\texttt{GAP}(\cdot)$ denotes global average pooling. With the guidance of displacements at both feature and matching levels, the fused representation $\boldsymbol{z}_{\texttt{fused}}$ can then be fed into an O2M Head to generate the final inflow and outflow counts. That is, Eq. (5) can be updated to

$$p^{ij}_{t-1,t} = \texttt{sigmoid}\left(\texttt{MLP}(\boldsymbol{z}_{\texttt{fused}})\right)\,. \tag{15}$$

c) Displacement-Informed Optimal Transport Loss: Following CGNet [12], we employ the Optimal Transport (OT) loss $\mathcal{L}_{\texttt{OT}}$ for group-level contrastive learning. For appearance learning, we take the same cost matrix $\boldsymbol{C}_{\texttt{appear}} \in \mathbb{R}^{m\times n}$ as CGNet, which is computed by similarity matrices. For displacement learning, we extract the cross-frame block of the prior cost $\boldsymbol{\gamma}$ to generate the displacement cost matrix $\boldsymbol{C}_{\texttt{disp}} \in \mathbb{R}^{m\times n}$, defined by

$$\boldsymbol{C}_{\texttt{disp}} = 1 - \frac{\exp(\hat{\boldsymbol{\gamma}}_{ij})}{\sum_k \exp(\hat{\boldsymbol{\gamma}}_{kj}) + \sum_k \exp(\hat{\boldsymbol{\gamma}}_{ik}) - \exp(\hat{\boldsymbol{\gamma}}_{ij})}\,, \tag{16}$$

where $\hat{\boldsymbol{\gamma}}_{ij} = \frac{\boldsymbol{\gamma}_{ij}-\mu_{\boldsymbol{\gamma}}}{\sigma_{\boldsymbol{\gamma}}+\epsilon}$, $\mu_{\boldsymbol{\gamma}}$ and $\sigma_{\boldsymbol{\gamma}}$ are the mean and standard deviation of the prior cost $\boldsymbol{\gamma}$, ϵ is a smoothing term to avoid zero division, and i, j and k are indices. Finally, we arrive at the displacement-informed OT loss

$$\begin{aligned} \mathcal{L}_{\texttt{D-OT}} &= -\min_{\boldsymbol{\Pi}} \sum_{i=1}^{n}\sum_{j=1}^{m} \boldsymbol{C}_{ij}\boldsymbol{\Pi}_{ij}\,, \\ \boldsymbol{C} &= \lambda \boldsymbol{C}_{\texttt{disp}} + (1-\lambda)\boldsymbol{C}_{\texttt{appear}}\,, \end{aligned} \tag{17}$$

where $\boldsymbol{\Pi}$ is the matching matrix computed by Sinkhorn algorithm [49], and λ is a hyper-parameter. Notably, $\boldsymbol{C}_{\texttt{disp}}$ is not only designed for displacement prior but also enhances the discriminability of appearance features by pushing away representations of distant pedestrians and by pulling close those of proximity during OT solving process.

F. Loss Function

Besides displacement-informed OT loss, the cross entropy loss $\mathcal{L}_{\texttt{cls}}$ is used to supervise the training of OMPM. For training efficiency, we use Hungarian algorithm [42] to select candidates that contain both correct matching pairs and incorrect pairs, and pairwise matching pairs are then fed to a modulator and an MLP. Here, for a pedestrian in a frame, we group this pedestrian and its neighboring ones within a certain normalized distance, 0.2 for example (according to Fig. 2(b)), in adjacent frames. Then all pairwise matches within this group are considered positive matches, while all remaining ones are negative samples. Finally, the total loss $\mathcal{L}_{\texttt{total}}$ used by OMAN++ takes the form

$$\mathcal{L}_{\texttt{total}} = \mathcal{L}_{\texttt{D-OT}} + \mathcal{L}_{\texttt{cls}}\,. \tag{18}$$

V. Experiments

To quantitatively evaluate the effectiveness of OMAN++, we conduct extensive experiments on three standard VIC benchmarks and our WuhanMetroCrowd dataset.

A. Datasets and Metrics

Datasets. Following previous work [11], [14], we choose the SenseCrowd [15], CroHD [16] and MovingDroneCrowd [14] datasets, and we also compare methods that require no ID annotation on our WuhanMetroCrowd dataset. SenseCrowd consists of 634 videos, with head coordinates and ID annotations. CroHD is an MOT dataset with 9 videos (4 for training, and 5 for testing), involving crowded indoor and outdoor scenes. MovingDroneCrowd is a recent VIC dataset with ID annotations of 89 videos. For VIC, we transform the ID annotations into inflow/outflow labels.

Metrics. We use the standard VIC metric Weighted Relative Absolute Errors (WRAE) to quantify the results, defined by

$$WRAE = \sum_{i=1}^{K} \frac{T_i}{\sum_{j=1}^{K} T_j} \frac{|N_i - \hat{N}_i|}{N_i} \times 100\%\,, \tag{19}$$

where K represents the video number, T_i denotes the length of the i-th video sequence, N_i is the ground-truth pedestrian count of the i-th video, and $\hat{N}_i$ is the predicted one. Since VIC also closely relates to counting, generic counting metrics such as Mean Absolute Error (MAE) and Mean Squared Error (MSE) are reported as well.

B. Implementation Details

Training Details. For pedestrian localization, we adopt the framework of Point quEry Transformer (PET) [4]. For a fair comparison with CGNet [12], the ConvNext-S [50] is used to encode pedestrians into descriptors. The learning rate and weight decay are both set to $1e^{-4}$, λ is set as 0.1 and all other training details are kept the same as CGNet [12]. Our model is trained on 4 RTX 3090 GPUs with the batch size set to 4.

Testing Details. Following previous work [11], [14], the frame interval σ is set to 15 for SenseCrowd [15], 75 for CroHD [16], 4 for MovingDroneCrowd [14] dataset, and 1 for WuhanMetroCrowd dataset. Each pedestrian patch is cropped at a fixed size of 96×64 pixels, following CGNet [12].

Evaluation protocol. Results on SenseCrowd, CroHD, and MovingDroneCrowd are primarily reproduced by published works [14], [44] where available; while missing entries (e.g.,

TABLE II
RESULTS ON SENSECROWD DATASET. D0∼D4 REPRESENTS DIFFERENT CROWD DENSITIES, RANGING [0,50), [50,100), [100,150), [150,200) AND [200,+∞). 'TYPE' INDICATES METHOD CATEGORY. BEST PERFORMANCE IS IN **BOLDFACE**, AND SECOND BEST IS UNDERLINED

Method	Type	MAE↓	MSE↓	WRAE(%)↓	Density level (MAE↓)				
					D0	D1	D2	D3	D4
FairMOT [35]	MOT	35.4	62.3	48.9	13.5	22.4	67.9	84.4	145.8
HeadHunter-T [16]	MOT	30.0	60.6	44.9	8.9	17.3	56.2	96.4	131.4
MOTIP [41]	MOT	26.5	52.0	39.5	9.2	17.8	50.5	82.0	130.5
SambaMOTR [40]	MOT	24.8	48.5	36.2	8.4	16.0	45.2	78.5	122.0
LOI [6]	LOI	24.7	33.1	37.4	12.5	25.4	39.9	104.1	115.2
DRNet [11]	VIC	12.3	24.7	12.7	4.1	8.0	23.3	50.0	77.0
CGNet [12]	VIC	8.9	17.7	12.6	5.0	**5.8**	**8.5**	25.0	63.4
FMDC [44]	VIC	14.3	28.9	14.9	4.8	8.8	22.5	54.1	88.8
PDTR [13]	VIC	9.6	17.6	11.4	4.6	6.8	14.7	23.6	60.6
SDNet [14]	VIC	8.6	15.9	11.0	4.0	6.6	12.6	24.9	47.9
VIC-SSL [45]	VIC	**7.6**	**12.6**	**10.4**	3.6	6.7	14.8	15.4	**30.8**
OMAN (Ours)	VIC	8.3	15.4	11.1	3.9	6.2	13.2	20.3	48.2
OMAN++ (Ours)	VIC	7.7	14.6	**10.4**	**3.3**	6.3	12.3	**12.5**	47.9

TABLE III
RESULTS ON MOVINGDRONECROWD DATASET. D0∼D3 REPRESENTS DIFFERENT CROWD DENSITIES, RANGING [0,150), [150,300), [300,450) AND [450,+∞). 'TYPE' INDICATES METHOD CATEGORY. BEST PERFORMANCE IS IN **BOLDFACE**, AND SECOND BEST IS UNDERLINED

Method	Type	MAE↓	MSE↓	WRAE(%)↓	Density level (MAE↓)			
					D0	D1	D2	D3
ByteTrack [8]	MOT	153.2	227.6	63.8	83.4	24.0	325.0	441.3
BoT-SORT [32]	MOT	150.6	223.5	62.5	82.5	**22.0**	327.0	430.0
OC-SORT [33]	MOT	203.6	276.8	87.8	101.5	232.0	405.0	569.3
MOTIP [41]	MOT	128.4	198.5	53.2	70.2	26.8	280.5	388.5
SambaMOTR [40]	MOT	120.5	185.8	50.5	67.8	25.5	268.0	365.2
LOI [6]	LOI	241.8	337.9	99.6	110.1	294.5	467.6	719.3
DRNet [11]	VIC	81.1	126.3	33.4	28.7	129.9	217.1	246.7
CGNet [12]	VIC	66.1	110.4	29.2	25.9	111.0	144.0	199.0
FMDC [44]	VIC	120.3	183.6	48.8	61.7	75.7	54.9	411.1
SDNet [14]	VIC	41.0	58.3	19.3	23.7	79.8	41.2	102.9
OMAN (Ours)	VIC	37.9	61.5	18.9	**16.4**	31.0	52.0	129.0
OMAN++ (Ours)	VIC	**36.1**	**56.6**	**17.2**	25.4	73.0	**15.0**	**77.3**

FMDC on SenseCrowd) and all results on WuhanMetroCrowd are reproduced in this paper using publicly released code. All methods take the same train/val/test split on each dataset.

C. Experimental Results

Results on SenseCrowd. Results are reported in Table II. MOT and LOI methods almost fail on all density levels, and errors become even larger in denser scenes, while VIC methods significantly reduce the errors. The results show that OMAN++ surpasses the VIC baselines in general across all metrics, achieving 7.7 MAE, 14.6 MSE, and 10.4% WRAE. MAE, MSE and WRAE are 10.5%, 8.2%, and 5.5% lower than SDNet [14], while 7.2%, 5.2%, and 6.3% lower than OMAN [17]. Without any extra pretraining, OMAN++ exhibits comparable performance compared to VIC-SSL, indicating that the proposed priors provide complementary gains to self-supervised representations. Specifically, our model reports



Fig. 9. **Visualization of O2M matching.** Red dots indicate inflows/outflows, green dots indicate shared pedestrians, colorful circles represent different groups, and blue lines indicate inferred O2M matches. Errors mainly occur in occluded regions due to ambiguous appearance or poor detections, where our O2M paradigm relaxes the O2O matching to accommodate hard matches.

the best results on D0 and D3, and the second-best on D2 and D4, covering both the sparsest and the densest scenes. The results demonstrate that, with a deeper insight of VIC, the one-to-many matching and the motion awareness of OMAN++ benefit matching on different density levels, especially in dense scenes. In particular, the gains at D3 and D4 over SDNet can be attributed to the O2M matching, which captures group-level context that is more robust than the cross-frame attention of SDNet under dense occlusion; the displacement prior further disambiguates visually similar pedestrians according to spatial temporal cues. In addition, to empirically validate the social grouping prior, Fig. 9 demonstrates that undetected, occluded or wrongly identified pedestrians are successfully matched to visible group members; the comparison between the qualitative results of O2O and O2M matching (with the same locator) confirm that group-level context indeed compensates for missing appearance cues.

Results on MovingDroneCrowd. As shown in Table III, OMAN++ also outperforms all other VIC methods on MovingDroneCrowd dataset. OMAN++ achieves the lowest MAE, MSE, WRAE, which are 12.0%, 2.9% and 10.9% lower than SDNet [14], and are 4.7%, 8.0% and 9.0% lower than OMAN [17]. Among the four density levels, OMAN++ outperforms on two most crowded scenes, D2 and D3, where the displacement prior effectively constrains matching to spatially nearby candidates in structured drone-view motion fields. However, we also observe that, compared with OMAN (OMAN++ without the displacement prior) and some MOT methods, OMAN++ exhibits a noticeable performance degradation on D1. The decline may be largely due to the limited test data on D1 (only one single video sequence) and the small motion of pedestrians. This suggests that the displacement prior, while broadly beneficial, can hurt performance in low-motion, moderate-density scenes where displacements are small and spatially ambiguous, causing the prior cost to introduce mild noise rather than useful signal.

Results on CroHD. Experimental results on the CroHD

TABLE IV
RESULTS ON CROHD DATASET. CROHD11∼CROHD15 REPRESENTS THE FIVE SCENES, WITH 133, 737 ,734, 1040, AND 321 AS GROUND-TRUTH COUNTS, RESPECTIVELY. VALUES IN '( )' REPRESENTS THE DEVIATION FROM GROUND-TRUTH COUNTS, AND '–' IS DUE TO UNRELEASED CODE. 'TYPE' INDICATES METHOD CATEGORY. BEST PERFORMANCE IS IN **BOLDFACE**, AND SECOND BEST IS UNDERLINED

Method	Type	MAE↓	MSE↓	WRAE(%)↓	MAE↓ on the five testing scenes				
					CroHD11 133	CroHD12 737	CroHD13 734	CroHD14 1040	CroHD15 321
FairMOT [35]	MOT	256.2	300.8	34.1	144 (11)	1164 (427)	1018 (284)	632 (408)	472 (1000)
HeadHunter-T [16]	MOT	253.2	351.7	32.7	198 (65)	636 (101)	219 (515)	458 (582)	**324 (3)**
ByteTrack [8]	MOT	213.2	340.2	30.8	160 (27)	**761 (24)**	1467 (733)	897 (143)	460 (139)
MOTIP [41]	MOT	185.3	268.5	25.8	175 (42)	850 (113)	1200 (466)	820 (220)	410 (89)
SambaMOTR [40]	MOT	175.8	255.0	24.2	172 (39)	845 (108)	1180 (446)	850 (190)	405 (84)
LOI [6]	LOI	305.0	371.1	46.0	72.4 (60)	493.1 (243)	275.3 (458)	409.2 (630)	189.8 (131)
DRNet [11]	VIC	141.1	192.3	27.4	164.6 (31)	1075.5 (338)	752.8 (18)	784.5 (255)	382.3 (61)
CGNet [12]	VIC	75.0	95.1	14.5	– (7)	– (72)	– (14)	– (144)	– (138)
FMDC [44]	VIC	54.2	61.7	10.7	138.9 (5)	664.3 (73)	818.0 (84)	1005.8 (35)	394.9 (73)
PDTR [13]	VIC	60.6	73.7	12.7	109 (24)	678 (59)	**729 (5)**	935 (105)	431 (110)
SDNet [14]	VIC	128.9	160.0	19.7	103.0 (30)	801.9 (64)	577.2 (157)	740.8 (299.2)	414.8 (93)
VIC-SSL [45]	VIC	40.6	50.9	8.8	-	-	-	-	-
OMAN (Ours)	VIC	38.6	50.5	8.3	142 (9)	636 (101)	715 (19)	1003 (37)	348 (27)
OMAN++ (Ours)	VIC	**35.4**	**48.8**	**7.9**	**138 (5)**	639 (98)	698 (36)	**1048 (8)**	291 (30)

TABLE V
RESULTS ON WUHANMETROCROWD DATASET. PLATFORM, TRANSFER, FARE GATE, LOBBY, ESCALATOR, AND EXIT/ENTRANCE REPRESENT SIX TYPICAL SCENES IN TEST SUBSETS. 'TYPE' INDICATES METHOD CATEGORY. BEST PERFORMANCE IS IN **BOLDFACE**, AND SECOND BEST IS UNDERLINED

Method	Type	MAE↓	MSE↓	WRAE(%)↓	WRAE(%)↓ on different scenes					
					Platform	Transfer	Fare Gate	Lobby	Escalator	Exit/Entrance
ByteTrack [8]	MOT	225.4	467.2	48.1	43.9	58.0	20.4	90.3	78.2	57.7
OC-Track [33]	MOT	288.8	550.2	64.6	48.9	80.2	55.6	96.1	87.7	72.0
CGNet [12]	VIC	164.0	344.6	35.0	21.2	60.9	39.2	48.7	46.6	42.8
SDNet [14]	VIC	151.8	415.4	31.1	22.7	**10.7**	29.2	67.9	71.9	**10.8**
OMAN (Ours)	VIC	130.3	257.3	34.4	48.5	43.1	18.8	63.2	40.2	13.1
OMAN++ (Ours)	VIC	**45.4**	**62.3**	**18.1**	**15.9**	16.9	**12.5**	**34.4**	**11.3**	22.8

dataset are shown in Table IV. We observe that our proposed OMAN++ is also the best performing one. Generally, MAE, MSE and WRAE are respectively reduced by 8.3%, 3.4%, and 4.8% when compared to OMAN [17], and reduced by 34.7%, 20.9%, and 26.2% when compared to FMDC [44]. Even against VIC-SSL [45] that benefits from additional self-supervised pretraining, OMAN++ achieves 12.8% lower MAE and 10.2% lower WRAE. CroHD11∼CroHD15 represent five different scenes with different densities, while the differences between predicted and true values in different scenes are generally small, with the lowest MAE on CroHD11 and CroHD14, indicating that our model is robust to various densities of the pedestrian. The consistent gains over CGNet on CroHD can be attributed to the displacement prior, which effectively disambiguates visually similar pedestrians in the structured indoor environments of CroHD, where motion patterns are regular and predictable.

Results on WuhanMetroCrowd. To better demonstrate the superiority of OMAN++ in crowded and congested scenes, we conduct a comparison on our proposed WuhanMetroCrowd dataset. Due to the lack of identity annotation, we exclude two identity-based approaches (DRNet [11] and FMDC [44]) and the unreleased PDTR [13], and we conduct comprehensive evaluations on WuhanMetroCrowd dataset (shown in Table V). Generally, OMAN++ demonstrates significant improvements over SDNet [14], achieving 70.1%, 85.0% and 41.8% reductions in MAE, MSE and WRAE, respectively. Compared to OMAN [17], OMAN++ still reduces MAE, MSE and WRAE by 65.2%, 75.8% and 47.4%. Particularly, OMAN++ also achieves the lowest WRAE across four scenes, including platform, fare gate, lobby and escalator, and the second lowest WRAE in transfer. We observe that all the approaches fail in lobby with a WRAE even above 30%, but OMAN++ lower the WRAE by 29.4% $\sim$ 64.2%. We further attribute the failure in lobby to two challenges: i) the O2M matching degrades because even group-level appearance cues become ambiguous when crowd density causes near-total overlap between individuals; and ii) the displacement prior becomes less discriminative because all pedestrians tend to exhibit highly similar motion pattern, with little useful information for discriminating inflows/outflows. Considering that lobby is one of the densest scenes for its largest capacity, OMAN++ indeed shows a rela-



Fig. 10. **Qualitative comparisons on the WuhanMetroCrowd dataset.** In each image pair, green dots on the left sub-image represent locating results of reference pedestrians in the previous frame; while in the current frame on the right, green dots denote 'pedestrian' class, red dots represent 'inflow' class, black rectangles are masked regions, and blue circles highlight the failure cases of matching according to the ground-truth labels. The locating results for CGNet are kept the same with the ones for OMAN++.

TABLE VI
CROSS-DATASET EVALUATION (SENSECROWD → WUHANMETROCROWD). ALL METHODS ARE TRAINED ON SENSECROWD AND DIRECTLY TESTED ON WUHANMETROCROWD WITHOUT FINE-TUNING. 'TYPE' INDICATES METHOD CATEGORY. BEST PERFORMANCE IS IN **BOLDFACE**, AND SECOND BEST IS UNDERLINED

Method	Type	MAE↓	MSE↓	WRAE(%)↓
MOTIP [41]	MOT	230.5	462.8	51.2
SambaMOTR [40]	MOT	222.8	448.5	49.5
LOI [6]	LOI	298.5	565.2	70.5
DRNet [11]	VIC	195.2	388.5	41.5
CGNet [12]	VIC	228.5	445.2	48.5
FMDC [44]	VIC	258.2	498.5	54.2
SDNet [14]	VIC	215.8	428.5	43.8
OMAN (Ours)	VIC	<u>168.5</u>	<u>345.8</u>	<u>37.2</u>
OMAN++ (Ours)	VIC	**142.5**	**282.4**	**33.5**

tively great adaptability to crowds. Notably, despite the strong results, the remaining performance gap highlights both the challenging nature of our WuhanMetroCrowd benchmark and opportunities for future improvement in VIC. To illustrate this, we provide some challenging cases in Fig. 10. We attribute the failures mainly to two typologies: (1) **Extreme density with near-total occlusion** (primarily in Lobby and dense Platform scenes), where neither appearance nor displacement provides sufficient discriminative signal; and (2) **Small, fast-moving pedestrians** at scene boundaries, where the locator itself produces unreliable detections, propagating errors into the matching stage.

Cross-Dataset Generalization. To further assess the generalization capability of OMAN++, we conduct a cross-dataset evaluation by training all methods on SenseCrowd but testing on WuhanMetroCrowd without fine-tuning (Table VI). All methods suffer a noticeable performance drop compared to their same-dataset counterparts (Table V), primarily due to the domain gap in crowd density, camera perspective, and scene layout between the two datasets. Nevertheless, OMAN++ remains the best-performing method, achieving a 34.0% lower MAE and a 23.5% lower WRAE, compared to SDNet, respectively. Notably, the relative advantage of OMAN++ over other VIC methods is also pronounced in the cross-dataset setting, suggesting that physical motion constraints provide some dataset-agnostic cues. This result also highlights the value of WuhanMetroCrowd as a more challenging and diverse benchmark for evaluating VIC generalization.

D. Ablation Study

All ablation experiments in this section are conducted on the SenseCrowd dataset.

Impact of Proposed Modules. Here we ablate our approach to justify the contribution of the social grouping prior in Table VII(a) and of the displacement prior in Table VII(b) and Table VII(c). One can observe that, both the priors contribute to all the three main improvements. We first notice that precision dropped when we apply only DASA without the supervision of $\mathcal{L}_{\mathtt{D-OT}}$. We attribute it mainly to insufficient supervision of prior embedding $\mathcal{P}$, due to $\mathcal{P}$ is only indirectly supervised by appearance-based constraints. We also observe that, besides the constraints on the learning of the displacement prior, the prior cost can also guide the model to filter out some errors due to the slight improvement when only $\mathcal{L}_{\mathtt{D-OT}}$ is applied. Finally, the integration of both priors yields synergistic performance gains, demonstrating the effectiveness of our cohesive framework. We also note that the individual contribution of Displacement Modulator appears modest in Table VII(b) (without $\mathcal{L}_{\mathtt{D-OT}}$), where the prior embedding $\mathcal{P}$ is only indirectly supervised by appearance-based constraints, limiting its expressiveness. The synergistic gain is clearly visible in Table VII(c) once $\mathcal{L}_{\mathtt{D-OT}}$ provides direct supervision.

Fusion Approach of Displacement Prior. To evaluate the effectiveness of our proposed displacement-aware modulation for OMPM module, we compare the modulation with directly concatenating the displacement prior embeddings with appearance features. The comparison result in Table VII(g) demonstrates that our coarse-to-fine modulation achieves a better balance between motion and appearance. It is mainly due to that concatenation introduces feature redundancy and conflicts, while our modulation adaptively recalibrates the appearance features guided by the geometric prior, thereby enhancing the performance of O2M matching.

TABLE VII
ABLATION STUDY ON SENSECROWD

(a) **Impact of social grouping prior**. We evaluate each module based only on social grouping prior without the guidance of displacement prior. ICG and OMPM finally forms the baseline OMAN

ICG	OMPM	MAE↓	MSE↓	WRAE(%)↓
		16.9	27.2	22.0
✓		9.6	19.8	12.9
	✓	8.8	16.4	11.8
✓	✓	8.3	16.0	11.4

(b) **Impact of spatial-temporal displacement prior WITHOUT** $\mathcal{L}_{\mathtt{D-OT}}$. We encapsulate the prior into the model via DASA and modulation. The baseline is OMAN without $\mathcal{L}_{\mathtt{D-OT}}$

DASA	Modulator	MAE↓	MSE↓	WRAE(%)↓
		8.3	16.0	11.4
✓		8.8	16.7	11.9
	✓	8.4	15.2	11.1
✓	✓	8.4	16.5	11.0

(c) **Impact of spatial-temporal displacement prior WITH** $\mathcal{L}_{\mathtt{D-OT}}$. We justify DASA and the modulator under the supervision of D-OT loss. The baseline is OMAN with $\mathcal{L}_{\mathtt{D-OT}}$

DASA	Modulator	MAE↓	MSE↓	WRAE(%)↓
		8.3	16.2	11.2
✓		8.2	16.0	11.0
	✓	7.9	14.8	10.8
✓	✓	7.7	14.6	10.4

(d) **Impact of the threshold** η

Threshold	MAE↓	MSE↓	WRAE(%)↓
0.0	8.4	16.2	11.5
0.1	7.8	14.5	10.2
0.2	7.7	14.6	10.4
0.3	8.0	14.8	10.7
0.5	8.4	15.5	11.9

(e) **Impact of localization error**

Error Rate	MAE↓	MSE↓	WRAE(%)↓
0.0% (Oracle)	7.6	13.9	10.1
10.0%	7.9	14.2	10.3
20.0%	8.2	15.2	10.7
30.0%	9.8	18.2	12.7
40.0%	17.4	28.5	19.4
13.9% (Ours)	7.7	14.6	10.4

(f) **Efficiency of different methods**

Method	Counter	FPS	#Param.
DRNet [11]	Density Map	7.01	18.76M
CGNet [12]	- (Localization)	14.45	49.45M
FMDC [44]	Density Map	7.49	28.30M
SDNet [14]	Density Map	2.46	31.46M
OMAN (Ours)	Localization	11.42	79.91M
OMAN++ (Ours)	Localization	7.86	81.20M

(g) **Fusion of displacement prior**

Approach	MAE↓	MSE↓	WRAE(%)↓
Concatenation	8.5	16.6	10.8
Modulation (OMAN++)	7.7	14.6	10.4

(h) **Impact of prior cost**

Approach	MAE↓	MSE↓	WRAE(%)↓
Displacement	8.6	17.1	10.9
Prior Cost (OMAN++)	7.7	14.6	10.4

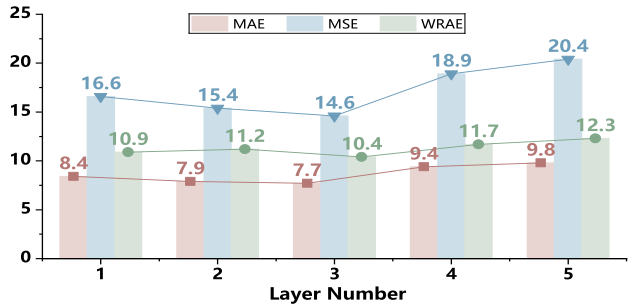
(i) **Impact of interval** σ. D0∼D4 represents different crowd densities, ranging [0,50), [50,100), [100,150), [150,200) and [200,+∞). Best performance is in **boldface**, and second best is underlined

σ	MAE↓	MSE↓	WRAE(%)↓	Density level (MAE↓)				
				D0	D1	D2	D3	D4
5 (1s)	12.7	20.8	18.6	6.7	9.8	31.1	38.4	**33.6**
10 (2s)	**7.6**	**13.5**	11.3	4.2	**5.9**	13.3	15.6	36.4
15 (3s)	7.7	14.6	**10.4**	**3.3**	6.3	**12.3**	**12.5**	47.9
20 (4s)	10.5	19.1	13.8	4.1	9.4	12.6	21.3	68.5
25 (5s)	12.1	21.0	15.7	4.3	11.3	16.6	27.1	71.8

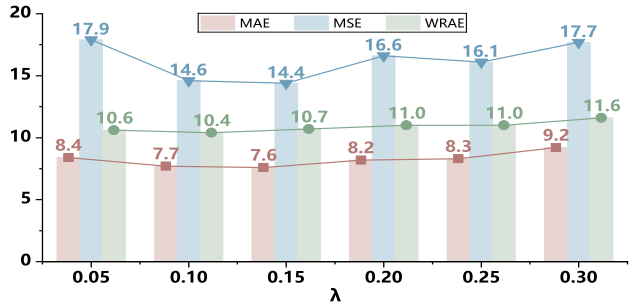
Performance with Different Layer Numbers. Since we implement our O2M head by a simple MLP, we conduct an experiment on the SenseCrowd dataset to determine the optimal layer configuration. As shown in Fig. 11(a), the performance is comparable with state-of-the-art methods with a shallow network, while too many layers can lead to overfitting, and the MAE, MSE, and WRAE all reach the minimum when the number of MLP layers is equal to three.

Impact of λ. In the initial stage of training, the loss for appearance should take the dominance to establish a coarse matching, because motion can only be learned after preliminary associations are formed. Subsequently, the constraint of displacement should be gradually enhanced to provide a fine-grained correspondence. Empirically, we find that a constant λ is sufficient because the appearance loss decays fast initially, allowing the displacement constraint to self-adjust its influence. We therefore show the optimal λ for OMAN++ is 0.1 in Fig. 11(b). We also observe that setting λ to 0.15 enables OMAN++ to achieve better performance in dense scenes (lower MAE and MSE), which is consistent with our analysis above, but it slightly sacrifices the performance in sparser scenes (higher WRAE).

Sensitivity to neighborhood threshold and η. We do not explicitly set η to a fixed value. Instead, for each pedestrian in the current frame, its positive group is defined as all pedestrians in the adjacent frame whose normalized distance falls within a threshold of 0.2; consequently, η is implicitly determined by the number of neighbours that satisfy this spatial criterion. The threshold of 0.2 is kept constant across all datasets. The configuration of the neighborhood threshold



(a) **Performance of O2M head with different layer numbers. The horizontal axis indicates the number of layers.**



(b) **Impact of λ. The horizontal axis indicates the value of λ.**

Fig. 11. **Impact of parameters on SenseCrowd.**



Fig. 12. **Qualitative analysis.** Each column shows two frame pairs from SenseCrowd, MovingDroneCrowd, and CroHD, including both high-angle surveillance and aerial drone viewing conditions. Red dots indicate inflows/outflows, green dots indicate shared pedestrians, and blue circles highlight false matches.

is mainly based on the statistical characteristics (the displacement of pedestrians is usually limited to within 0.2, which can be observed in Fig. 2); to verify our choice of 0.2, Table VII(d) reports the performance of OMAN++ with the threshold ranging from 0 to 0.5, where OMAN++ shows good robustness when the threshold is in the range of 0.1 and 0.3.

Impact of Locators. To isolate how localization quality affects VIC, we replace PET predictions with ground-truth locations perturbed by varying noise including random missing, duplicated, or displaced detections on SenseCrowd. As shown in Table VII(e), clean GT locations yield an MAE of 7.6, while the actual PET locator achieves MAE 7.7 and WRAE 10.4%, falling between the 10% and 20% synthetic noise rows and consistent with the oracle, confirming that synthetic noise faithfully predicts the correlation between localization and matching. Degradation is decent up to 30% error, thanks to the relaxed O2M matching and the displacement prior.

Efficiency Comparison. OMAN++ demonstrates significant advantages over other methods across computational efficiency (Table VII(f)). Note that CGNet [12] does not incorporate a locator. Despite higher parameters, our methods achieve higher inference speed beyond all density map-based methods through efficient resource utilization. OMAN++ processes frames in a coarse-to-fine manner and implements matching at the individual level rather than on the whole image, facilitating real-time employment in the real world.

E. Discussion

Analysis of prior cost. To justify our learnable prior cost over static metrics like Euclidean distance, we first depict the relationship between prior cost and displacement in Fig. 7. One can observe that prior cost tends to be more sensitive to displacement in x-axis than in y-axis, which simply verifies our grounds that OMAN++ models the anisotropic patterns of motion caused by camera projection, scene layout and pedestrian mobility that may induce axis-dependent statistics. Projecting the raw displacement prior into a learned latent space i) compensates for axis scaling and correlation; ii) allows non-linear sensitivity to displacement magnitudes; and iii) produces a displacement-aware term that fuses naturally with appearance similarity. To further demonstrate the effectiveness, we compare the results by replacing our prior cost with the displacement itself (Table VII(h)).

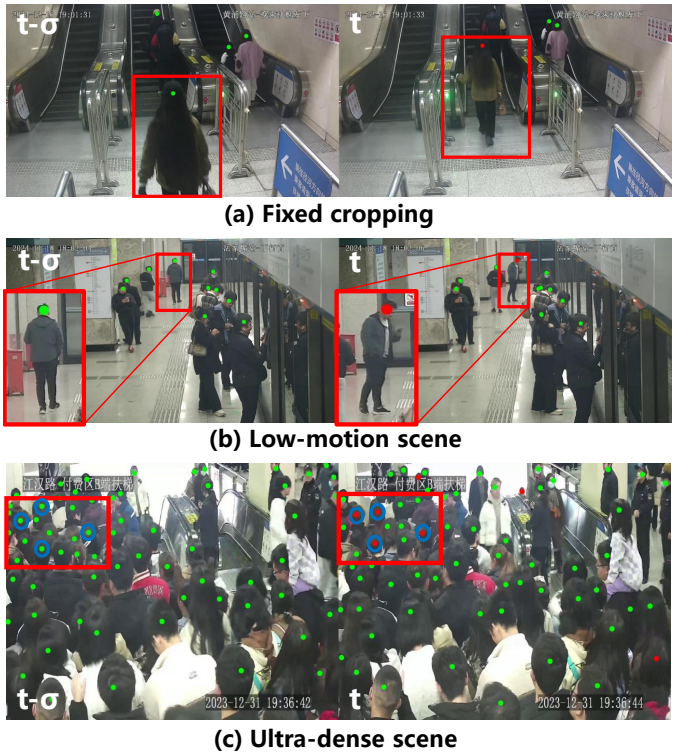


Fig. 13. **Failures on WuhanMetroCrowd.** (a) fixed cropping size which may sacrifice the robustness to scale variation (the two red bounding boxes take the same size, while the scale of the pedestrian increases); (b) low-motion scene where appearance can change significantly while displacement prior may also fail due to spatially ambiguous motion. (c) ultra-dense scene where both appearance and displacement cues are insufficient for accurate matching.

Impact of frame interval σ. For a fair comparison, we simply keep the same frame interval σ as existing works; but for a comprehensive understanding of the σ, we evaluate OMAN++ with different σ values in Table VII(i). The model achieves the lowest WRAE when σ is set to 3s, consistent with CGNet [12]. We also observe that in densest scenarios (D4), performance improves with smaller intervals, likely because dense crowds exhibit minimal structural change over shorter periods, making motion-wise correlation easier to capture.

Qualitative Analysis. Beyond the low-angle surveillance perspective featured in WuhanMetroCrowd, additional qualitative results in Fig. 12 on SenseCrowd, MovingDroneCrowd, and CroHD reveal distinct advantages and failure patterns: i)

on SenseCrowd, the high-angle surveillance perspective mitigates occlusion, and OMAN++ achieves strong performance with only minor mismatches that is mainly due to variable behavior patterns of a single person; ii) on MovingDroneCrowd, the model is robust in structured outdoor environments but struggles with low-resolution small targets that degrade both localization and matching; iii) on CroHD, OMAN++ tackles scale-consistent crowds well, yet occasionally underestimates inflows in tightly-packed regions where uniform motion patterns reduce the discriminability of the displacement prior. These observations suggest that, while the proposed priors generalize across viewing conditions in general, their effectiveness is also modulated by other factors including image resolution, crowd density, and motion homogeneity.

Limitations. Despite the strong results, several limitations deserve attention: i) *Fixed-scale cropping*: pedestrian patches are cropped at a fixed 96×64 pixels, which may be suboptimal in scenes with extreme scale variation (Fig 13(a)). ii) *Displacement prior in low-motion scenes*: as shown in Fig 13(b), the displacement prior can be counterproductive when pedestrian motion is small and spatially ambiguous; an adaptive motion-aware prior weight could address this. iii) *Ultra-dense scenes*: a significant performance gap persists in extremely dense scenes (Fig 13(c)), suggesting that current O2M and displacement-prior formulations are insufficient for extreme densities with various motion patterns; future work may explore crowd flows or multi-scale density modeling.

VI. Conclusion

In this work, we first construct a novel VIC benchmark, WuhanMetroCrowd, to fill current data gap of crowded scenarios. WuhanMetroCrowd features diverse density levels, long video sequences, dynamic pedestrian flows, and intensive occlusion. To further adapt VIC methods to dense crowds, we conclude our two observations into a social group prior and a spatial-temporal displacement prior. Based on the two informative priors, we introduce a simple but effective model called OMAN++ for VIC. We first implicitly generate token-to-token similarity context with an ICG module and implement O2M assignment by an OMPM module, to inform the group context. We then exploit the displacement prior with a DPI module, infusing the appearance and motion cues through a DASA block, a displacement-aware modulator, and a D-OT loss. Experiments on the three standard benchmarks and WuhanMetroCrowd dataset indicate state-of-the-art performance. Due to the simplicity of implementation and a fresh perspective, we believe OMAN++ will serve as a strong baseline for VIC with ample scope for improvement.

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References

[1] L. Liu, J. Chen, H. Wu, J. Zhen, G. Li, and L. Lin, "Physical-virtual collaboration modeling for intra- and inter-station metro ridership prediction," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 4, pp. 3377–3391, 2022. 1

[2] Y. Fang, B. Zhan, W. Cai, S. Gao, and B. Hu, "Locality-constrained spatial transformer network for video crowd counting," in *Int. Conf. Multimedia and Expo*, 2019, pp. 814–819. 1, 3

[3] X. Wu, B. Xu, Y. Zheng, H. Ye, J. Yang, and L. He, "Fast video crowd counting with a temporal aware network," *Neurocomputing*, vol. 403, pp. 13–20, 2020. 1, 3

[4] C. Liu, H. Lu, Z. Cao, and T. Liu, "Point-query quadtree for crowd counting, localization, and more," in *IEEE Int. Conf. Comput. Vis.*, 2023, pp. 1676–1685. 1, 3, 9

[5] Z. Ma and A. B. Chan, "Crossing the line: Crowd counting by integer programming with local features," in *IEEE Conf. Comput. Vis. Pattern Recognit.*, 2013, pp. 2539–2546. 1, 3

[6] Z. Zhao, H. Li, R. Zhao, and X. Wang, "Crossing-line crowd counting with two-phase deep neural networks," in *Eur. Conf. Comput. Vis.*, 2016, pp. 712–726. 1, 3, 10, 11, 12

[7] H. Zheng, Z. Lin, J. Cen, Z. Wu, and Y. Zhao, "Cross-line pedestrian counting based on spatially-consistent two-stage local crowd density estimation and accumulation," *IEEE Trans. Circuit Syst. Video Technol.*, vol. 29, no. 3, pp. 787–799, 2019. 1, 3

[8] Y. Zhang, P. Sun, Y. Jiang, D. Yu, F. Weng, Z. Yuan, P. Luo, W. Liu, and X. Wang, "ByteTrack: Multi-object tracking by associating every detection box," in *Eur. Conf. Comput. Vis.*, 2022, p. 1–21. 1, 3, 10, 11

[9] T. Zhou, Q. Ye, W. Luo, H. Ran, Z. Shi, and J. Chen, "Apptracker+: Displacement uncertainty for occlusion handling in low-frame-rate multiple object tracking," *Int. J. Comput. Vis.*, pp. 1–26, 2024. 1, 3

[10] P. Chu, J. Wang, Q. You, H. Ling, and Z. Liu, "TransMOT: Spatial-temporal graph transformer for multiple object tracking," in *Winter Conf. Appl. Comput. Vis.*, 2023, pp. 4859–4869. 1, 3, 7

[11] T. Han, L. Bai, J. Gao, Q. Wang, and W. Ouyang, "DR.VIC: Decomposition and reasoning for video individual counting," in *IEEE Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 3073–3082. 1, 2, 3, 9, 10, 11, 12, 13

[12] X. Liu, G. Li, Y. Qi, Z. Yan, Z. Han, A. van den Hengel, M.-H. Yang, and Q. Huang, "Weakly supervised video individual counting," in *IEEE Conf. Comput. Vis. Pattern Recognit.*, 2024, pp. 19 228–19 237. 2, 3, 5, 6, 9, 10, 11, 12, 13, 14

[13] R. Li, Y. Liu, H. Li, J. Li, and G. Lu, "Prototype-guided dual-transformer reasoning for video individual counting," in *ACM Int. Conf. Multimedia*, 2024, p. 10258–10267. 2, 3, 10, 11

[14] Y. Fan, J. Wan, T. Han, A. B. Chan, and A. J. Ma, "Video individual counting for moving drones," in *IEEE Int. Conf. Comput. Vis.*, 2025. 2, 4, 5, 9, 10, 11, 12, 13

[15] H. Li, L. Liu, K. Yang, S. Liu, J. Gao, B. Zhao, R. Zhang, and J. Hou, "Video crowd localization with multifocus gaussian neighborhood attention and a large-scale benchmark," *IEEE Trans. Image Proc.*, vol. 31, p. 6032–6047, 2022. 2, 5, 6, 9

[16] R. Sundararaman, C. De Almeida Braga, E. Marchand, and J. Pettré, "Tracking pedestrian heads in dense crowd," in *IEEE Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 3864–3874. 2, 3, 5, 6, 9, 10, 11

[17] X. Zhu, J. Xu, B. Wang, H. Dai, and H. Lu, "Video individual counting with implicit one-to-many matching," in *IEEE Int. Conf. Image Proc.*, 2025. 3, 6, 10, 11

[18] H. Idrees, I. Saleemi, C. Seibert, and M. Shah, "Multi-source multi-scale counting in extremely dense crowd images," in *IEEE Conf. Comput. Vis. Pattern Recognit.*, 2013, pp. 2547–2554. 3

[19] V. B. Subburaman, A. Descamps, and C. Carincotte, "Counting people in the crowd using a generic head detector," in *Int. Conf. Adv. Video and Signal-Based Surveillance*, 2012, pp. 470–475. 3

[20] D. B. Sam, S. V. Peri, M. N. Sundararaman, A. Kamath, and R. V. Babu, "Locate, size, and count: Accurately resolving people in dense crowds via detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 8, pp. 2739–2751, 2021. 3

[21] Y. Liu, M. Shi, Q. Zhao, and X. Wang, "Point in, box out: Beyond counting persons in crowds," in *IEEE Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 6462–6471. 3

[22] V. Lempitsky and A. Zisserman, "Learning to count objects in images," *Adv. in neural Inf. Process. Syst.*, vol. 23, 2010. 3

[23] W. Liu, M. Salzmann, and P. Fua, "Context-aware crowd counting," in *IEEE Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 5094–5103. 3

[24] Q. Song, C. Wang, Z. Jiang, Y. Wang, Y. Tai, C. Wang, J. Li, F. Huang, and Y. Wu, "Rethinking counting and localization in crowds: A purely point-based framework," in *IEEE Int. Conf. Comput. Vis.*, 2021, pp. 3345–3354. 3

[25] J. Cheng, H. Xiong, Z. Cao, and H. Lu, "Decoupled two-stage crowd counting and beyond," *IEEE Trans. Image Proc.*, vol. 30, pp. 2862–2875, 2021. 3

[26] Y. Cong, H. Gong, S.-C. Zhu, and Y. Tang, "Flow mosaicking: Real-time pedestrian counting without scene-specific learning," in *IEEE Conf. Comput. Vis. Pattern Recognit.*, 2009, pp. 1093–1100. 3

[27] A. B. Chan and N. Vasconcelos, "Counting people with low-level features and bayesian regression," *IEEE Trans. Image Proc.*, vol. 21, no. 4, pp. 2160–2177, 2012. 3

[28] T.-Y. Chen, C.-H. Chen, D.-J. Wang, and Y.-L. Kuo, "A people counting system based on face-detection," in *Int. Conf. Genetic Evol. Comput.*, 2010, pp. 699–702. 3

[29] B.-K. Dan, Y.-S. Kim, Suryanto, J.-Y. Jung, and S.-J. Ko, "Robust people counting system based on sensor fusion," *IEEE Trans. Consum. Electron.*, vol. 58, no. 3, pp. 1013–1021, 2012. 3

[30] Z. Ma and A. B. Chan, "Counting people crossing a line using integer programming and local features," *IEEE Trans. Circuit Syst. Video Technol.*, vol. 26, no. 10, pp. 1955–1969, 2016. 3

[31] S. K. Zhou, R. Chellappa, and B. Moghaddam, "Visual tracking and recognition using appearance-adaptive models in particle filters," *IEEE Trans. Image Proc.*, vol. 13, no. 11, pp. 1491–1506, 2004. 3

[32] N. Aharon, R. Orfaig, and B.-Z. Bobrovsky, "Bot-sort: Robust associations multi-pedestrian tracking," 2022, arXiv:2206.14651. 3, 10

[33] J. Cao, J. Pang, X. Weng, R. Khirodkar, and K. Kitani, "Observation-centric sort: Rethinking sort for robust multi-object tracking," in *IEEE Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 9686–9696. 3, 10, 11

[34] Z. Wang, L. Zheng, Y. Liu, Y. Li, and S. Wang, "Towards real-time multi-object tracking," in *Eur. Conf. Comput. Vis.* Springer, 2020, pp. 107–122. 3

[35] Y. Zhang, C. Wang, X. Wang, W. Zeng, and W. Liu, "FairMOT: On the fairness of detection and re-identification in multiple object tracking," *Int. J. Comput. Vis.*, vol. 129, no. 11, p. 3069–3087, 2021. 3, 10, 11

[36] J. Peng, C. Wang, F. Wan, Y. Wu, Y. Wang, Y. Tai, C. Wang, J. Li, F. Huang, and Y. Fu, "Chained-tracker: Chaining paired attentive regression results for end-to-end joint multiple-object detection and tracking," in *Eur. Conf. Comput. Vis.* Springer, 2020, pp. 145–161. 3

[37] X. Zhou, V. Koltun, and P. Krähenbühl, "Tracking objects as points," in *Eur. Conf. Comput. Vis.* Springer, 2020, pp. 474–490. 3

[38] T. Meinhardt, A. Kirillov, L. Leal-Taixe, and C. Feichtenhofer, "Trackformer: Multi-object tracking with transformers," in *IEEE Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 8844–8854. 3

[39] P. Sun, J. Cao, Y. Jiang, R. Zhang, E. Xie, Z. Yuan, C. Wang, and P. Luo, "Transtrack: Multiple object tracking with transformer," 2020, arXiv:2012.15460. 3

[40] M. Segu, L. Piccinelli, S. Li, Y.-H. Yang, L. Van Gool, and B. Schiele, "Samba: Synchronized set-of-sequences modeling for multiple object tracking," in *Int. Conf. Learn. Represent.*, vol. 2025, 2025, pp. 30 057–30 070. 3, 10, 11, 12

[41] R. Gao, J. Qi, and L. Wang, "Multiple object tracking as id prediction," in *IEEE Conf. Comput. Vis. Pattern Recognit.*, 2025, pp. 27 883–27 893. 3, 10, 11, 12

[42] H. W. Kuhn, "The hungarian method for the assignment problem," *Nav. Res. Logistics Quart.*, vol. 2, no. 1-2, pp. 83–97, 1955. 3, 9

[43] G. Peyré, M. Cuturi *et al.*, "Computational optimal transport: With applications to data science," *Found. Trends Mach. Learn.*, vol. 11, no. 5–6, p. 355–607, 2019. 3

[44] C.-L. Wan, F.-K. Huang, and H.-H. Shuai, "Density-based flow mask integration via deformable convolution for video people flux estimation," in *Winter Conf. Appl. Comput. Vis.*, 2024, pp. 6559–6568. 3, 9, 10, 11, 12, 13

[45] F.-K. Huang, B.-L. Huang, L.-W. Tsao, J.-C. Wu, H.-H. Shuai, and W.-H. Cheng, "Flowing crowd to count flows: A self-supervised framework for video individual counting," in *ACM Int. Conf. Multimedia*, 2025, pp. 8234–8243. 4, 10, 11

[46] Z. Wang, L. Xiao, Z. Cao, and H. Lu, "Vision transformer off-the-shelf: a surprising baseline for few-shot class-agnostic counting," in *AAAI Conf. Artif. Intell.*, 2024, p. 5832–5840. 7

[47] Y. Yan, J. Qin, B. Ni, J. Chen, L. Liu, F. Zhu, W.-S. Zheng, X. Yang, and L. Shao, "Learning multi-attention context graph for group-based re-identification," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 6, pp. 7001–7018, 2023. 7

[48] B.-W. Yin, J.-L. Cao, M.-M. Cheng, and Q. Hou, "Dformerv2: Geometry self-attention for rgbd semantic segmentation," in *IEEE Conf. Comput. Vis. Pattern Recognit.*, 2025, pp. 19 345–19 355. 8

[49] R. Sinkhorn and P. Knopp, "Concerning nonnegative matrices and doubly stochastic matrices," *Pacific J. of Math.*, vol. 21, pp. 343–348, 1967. 9

[50] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A convnet for the 2020s," in *IEEE Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 11 976–11 986. 9



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